

IBC-26A-00-01-3S

27 August 2026 17:39

IBC-26A-00-01-3S

IMPACT 2026 – Phase 00

Assignment 01

Train Shunting Puzzle – HYPER-SONIC (3S) Level

Document Control

Program Code : IBC-26A-00-01-3S

Batch : 26B

Level : 3S (HYPER-SONIC)

Start Date & Time : __27-02-2026 :: 5:45_____

End Date & Time : __29-02-2026 :: 9:00_____

1. Participant Brief

This is the HYPER-SONIC (3S) version of Assignment 01. You are required to design a measurable, efficient shunting algorithm. Correctness alone is insufficient. You must demonstrate optimization, modularity, structured design discipline, and publish verifiable evidence.

2. System Layout (Five-Track Hub System)

- Tracks: A, B, C, D, and E.
- Track E contains Engine EE.
 - Tracks A, B, C, D contain mixed rakes.
 - Track E connects independently to A, B, C, D via intersections EA, EB, EC, ED.
 - All tracks have unlimited capacity.
 - Final required parking locations:
 - AG (All-GREEN) must be on Track A.
 - AB (All-BLUE) must be on Track B.
 - AR (All-RED) must be on Track C.
 - Track D must end empty.
 - Engine EE must end on Track E.

3. Initial Rolling Stock

- Track A: Rake PP – 20 wagons (mixed GREEN/BLUE/RED)
- Track B: Rake QQ – 24 wagons (mixed GREEN/BLUE/RED)
- Track C: Rake RR – 18 wagons (mixed GREEN/BLUE/RED)
- Track D: Rake SS – 27 wagons (mixed GREEN/BLUE/RED)

Wagons are unordered within each rake.

4. Objective

Using Engine EE, reorganize wagons to form exactly three uniform-color rakes:

- AG – all GREEN wagons
- AB – all BLUE wagons
- AR – all RED wagons

All wagons must be classified. No wagon may be lost or duplicated.

After completion, display:

- Total GREEN wagons in AG
- Total BLUE wagons in AB
- Total RED wagons in AR

5. Movement and Cost Model (Mandatory)

- Engine movement cost: ₹500 per meter (forward or backward).
- Engine EE is 100 meters away from any intersection (EA/EB/EC/ED).
- Each rake is parked 200 meters away from its nearest intersection.
- Coupling/decoupling time is considered zero cost.
- Only physical engine movement contributes to cost.

You must compute and publish:

- 1) Total meters traveled by EE.
- 2) Total fuel cost = Total meters × 500.

6. Operational Rules (Clarified for Robustness)

1. Engine may carry a maximum of 10 wagons per trip.
2. Wagons moved in one trip must be contiguous within the source rake.
3. You may reorganize wagons within a rake by shunting.
4. Track D must be fully emptied by the end.
5. Final uniform rakes must be contiguous (no mixing allowed).
6. A wagon once placed into AG, AB, or AR cannot be reclassified.

7. Optimization Requirement

Primary metric: Minimize total engine meters traveled.

Secondary metric (tie-breaker): Minimize number of engine trips.

Your algorithm must demonstrate logical batching or optimization strategy.

8. Mandatory Structured Modules (3R Enforcement)

Your structured flowchart must contain reusable modules including:

- SCAN_RAKE()
- SELECT_SOURCE()
- SELECT_DESTINATION(color)
- MOVE_ENGINE(from, to) – must update cumulative meters
- SHUNT_BLOCK(source, destination)
- UPDATE_STATE()
- TERMINATION_CHECK()

Each module must be one-entry-one-exit and reusable without modification.

9. Evidence Package (Mandatory for 3S Qualification)

You must submit:

- 1) Master structured flowchart (top level).
 - 2) Exploded modular flowcharts.
 - 3) State distribution table (color counts per track at key checkpoints).
 - 4) Engine movement ledger (each movement + cumulative meters).
 - 5) Final results summary (AG, AB, AR counts + total meters + total cost).
 - 6) Efficiency justification note (1–2 pages).
- Correctness of final configuration.
 - Structural rigor and 3R compliance.
 - Optimization quality.
 - Evidence clarity and traceability.

End of HYPER-SONIC (3S) Assignment Document